

Do Simulated Conversations Reproduce Real Agent Failures?

A Scorer-Parity Comparison and Scale Study of the Samsung WhatsApp Support Agent

Eduardo Fernandez Salazar

Imperial College London · MSc Artificial Intelligence · Pulpoo / Samsung Polanco

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Abstract

Synthetic adversarial testing of conversational agents is widely used, but whether the failures it produces correspond to the failures that occur in real production is rarely measured. We compare 1,299 real Spanish WhatsApp support conversations from a deployed Samsung service agent (376 failures under a rule-based, LLM-free scorer) against up to 2,560 simulated conversations generated by LLM personas talking to a *verbatim* copy of the same production agent running on a fully seeded fake database. Both corpora are labelled by the *identical* scorer and projected onto a frozen 16-category failure taxonomy, giving scorer parity by construction. The simulation reproduces 5 of 7 reachable real failure categories (71%), including the dominant production failure mode (loops/stalls: 68% of real failures, 81% of simulated), at a similar overall failure rate (21% vs 29%). A seven-batch scale study ($N = 10$ to $N = 1000$) shows category coverage saturates at $N \approx 50$ and never improves thereafter: the two absent categories (comprehension, data-gap failures — jointly present in over half of real failures) are *structurally* unreachable under current user simulators and test-data seeding, not budget-limited. The Jensen–Shannon divergence between real and simulated failure distributions is scale-invariant at ≈ 0.24 bits (sampling noise floor: 0.005), demonstrating that increasing simulation volume buys statistical precision and failure volume, but not new failure kinds. We conclude that simulation power is gated by simulator design — persona realism and backend-data imperfection — rather than by scale, and quantify exactly which production failure modes remain out of reach.

1 Introduction

Conversational agents deployed to customers fail in ways their test suites did not anticipate. The dissertation this report belongs to (*Closing the Loop Between Synthetic Agent Testing and Production Reality*) asks whether synthetic testing predicts production failures. Earlier *static-mode* analysis answered a design-level version of that question: test suites generated without production knowledge *target* only a small fraction of observed production failure categories (recall 0.068). This report answers the stronger, behavioural version:

When simulated users converse with the real agent code, do the failures that actually emerge match the failures observed in production — in kind, in frequency, and in mechanism? And how does that correspondence grow with the number of simulations?

This became answerable when the production agent was reproduced *verbatim* inside a controlled simulation: identical prompts, LangGraph structure, and guardrails, with only the database layer replaced by a fully specified in-memory fake. The agent cannot distinguish the simulation from production; its only live dependency is its own LLM calls.

Contributions.

1. A **scorer-parity methodology**: both corpora are labelled by the same rule-based, LLM-free failure scorer and projected onto a frozen shared taxonomy, eliminating judge-model bias from the comparison itself (§3).
2. A **correspondence result**: 5/7 reachable real failure categories reproduced, with the production-dominant loop/stall mode reproduced as the simulation-dominant mode (§4).
3. A **scale study** across seven independent batches ($N = 10 \dots 1000$; 2,560 conversations) showing coverage saturation at $N \approx 50$ and scale-invariant distributional divergence (§5).
4. A **reachability analysis** separating structurally unreachable failure modes from budget-limited ones, with the mechanisms identified (§6).

2 Data

2.1 Real corpus

The production corpus contains **1,299 conversations** (24,537 messages) between real customers and the deployed Samsung WhatsApp support agent, exported once and anonymised by a three-pass pipeline (regex, Spanish NER, brand scrub); all experiments run offline on the anonymised corpus. Structured operational signals accompany each conversation: escalation records, human takeovers, message delivery statuses, intent/confidence telemetry, and timestamps (291 escalations; 22.4%).

2.2 Simulated corpora

The simulation target is the production agent copied verbatim into a sandbox (`samsung-live-agent/`): byte-identical prompts, the same graph and guardrails, an in-memory Supabase-compatible fake database seeded with one fully specified fictitious customer (two service orders exercising both warranty states, a disclosure-gated order, CRM memory, and escalation rules). Simulated users are LLM personas (Claude Haiku, Spanish) generated by the platform’s Phase B pipeline from the agent’s own structure (MAP-Elites persona library, scenario catalogue with adversarial variants), equipped with the fake customer’s identifying context. Conversations run over HTTP with per-test session isolation and per-batch database resets. The primary corpus for the distribution comparison pools 390 conversations from three independent runs; the scale study adds 2,560 more in seven independent batches.

3 Methodology

3.1 Scorer parity

The methodological core is that *the same* failure detector labels both corpora. The production scorer is rule-based and LLM-free: it reads only operational signals — escalations, explicit human requests (Spanish pattern lexicon), message counts, verbatim customer repetition, verbatim agent repetition, delivery failures, expiry without resolution, frustration lexicon, and customer pushback wording. It emits a weighted failure score and one or more of eight production failure categories. A conversation is a failure when its score is ≥ 3 *and* at least one category triggered — the same ground-truth criterion used for the real corpus (376/1,299 failures).

Simulated conversations are adapted to the production message schema by a thin, faithful mapping (user→customer, agent→ai_agent, tool calls preserved, persona abandonment → expiry, filed escalations → escalation records). Telemetry that does not exist in simulation (intent detection, confidence scores, delivery statuses) is left absent, exactly as it would be for a production conversation missing those fields. No signal is invented for either side.

3.2 Shared taxonomy and projection

Both category vocabularies project onto a frozen 16-category shared taxonomy¹ through total, import-time-checked mappings. All distributional comparisons are expressed in that taxonomy.

3.3 Distributional comparison

Let P and Q be the normalised distributions of category labels among failed conversations in the real and simulated corpora, over the pre-registered *reachable support* (§3.4). We report the Jensen–Shannon divergence

$$\text{JSD}(P \parallel Q) = \frac{1}{2} D_{\text{KL}}(P \parallel M) + \frac{1}{2} D_{\text{KL}}(Q \parallel M), \quad M = \frac{1}{2}(P + Q), \quad (1)$$

in bits (symmetric, bounded $[0, 1]$), with 95% bootstrap confidence intervals (1,000 conversation-level resamples). Two anchors make the number interpretable: a **noise floor** (mean JSD between random halves of the real corpus itself, 0.0050; p95 0.0108) and the **real-vs-uniform** JSD (0.2014). Per-category rates carry Wilson 95% intervals.

3.4 Reachability pre-registration

Before running, each production category was classified by whether the simulator *could* produce it: **delivery_infra** is *unreachable* (no WhatsApp transport layer exists in simulation); **comprehension** is *partial* (the customer-repetition signal exists, but intent/confidence telemetry does not); **data_gap** is *partial* (a single fully seeded customer leaves little room for missing backend data); the remaining five are fully reachable. Two failure-mode predictions were pre-registered from the literature: simulated users are systematically over-cooperative and rarely repeat themselves verbatim or express frustration (Lost in Simulation; Synthetic Users, Real Differences), and a complete test database suppresses data-gap failures.

3.5 Scale study design

Seven **independent** batches of $N \in \{10, 50, 100, 200, 400, 800, 1000\}$ conversations were executed sequentially against the live agent (six workers; fresh database reset per batch; deterministic seeded suites: for $N \leq 150$ a seeded sample of the 150-test suite, above that the full suite plus seeded resampled duplicates with fresh test identities). Per batch we measure failure count and rate, distinct categories found, coverage of reachable real categories, JSD to the real distribution, wall time, and persona cost; the same metrics are computed cumulatively over the pooled batches (2,560 conversations).

4 Results: correspondence at fixed scale

4.1 Corpus-level rates

Under the identical scorer and threshold, the two corpora fail at similar rates (Table 1): 28.9% of real conversations versus 21.0% of simulated ones.

Table 1: Corpus-level failure rates under the identical rule-based scorer (min score = 3).

	Conversations	Failures	Failure rate
Real (production)	1,299	376	28.9%
Simulated (pooled runs)	390	82	21.0%

¹Version 1.0-frozen-2026-07-14; frozen mid-campaign so that precision/recall comparisons stay meaningful against a fixed vocabulary.

4.2 Reachability matrix

Table 2 is the study’s central artefact: for each production category, whether the simulator could produce it, and whether it did.

Table 2: Reachability matrix (production vocabulary; counts are failures carrying each label).

Production category	Reachability	Real	Sim	Status
comprehension	partial	153	0	missed
resolution	reachable	221	22	found
data_gap	partial	48	0	missed
loop_stall	reachable	256	66	found
delivery_infra	unreachable	14	0	unreachable
missed_escalation	reachable	11	19	found
silent_abandonment	reachable	9	10	found
hallucination	reachable	13	2	found

Coverage: 5 of 7 reachable real categories are reproduced in simulation (71%). Both misses were pre-registered predictions, observed exactly as stated (§3.4, discussed in §6).

4.3 Category distributions

Figure 1 and Table 3 compare per-category rates among failures. The correspondence at the top is striking: **the dominant production failure mode is also the dominant simulated failure mode**. Loops/stalls (`infinite_loop`) account for 68.1% of real failures and 80.5% of simulated ones. Escalation-related failures rank second on both sides when `resolution_failure` and `escalation_failure` are viewed together (real 58.8%+2.9%; simulated 26.8%+23.2%), though the simulation splits mass differently between them: simulated frustration wording more often precedes a *non*-escalated ending, whereas real frustrated customers usually get (or force) the hand-off.

Table 3: Per-category rates among failed conversations (shared taxonomy; Wilson 95% CIs).

Category	Real ($n=376$ failures)		Simulated ($n=82$)	
	Rate	95% CI	Rate	95% CI
<code>infinite_loop</code>	68.1%	[63.2, 72.6]	80.5%	[70.6, 87.6]
<code>resolution_failure</code>	58.8%	[53.7, 63.6]	26.8%	[18.4, 37.3]
<code>comprehension_failure</code>	40.7%	[35.8, 45.7]	0.0%	—
<code>data_gap</code>	12.8%	[9.8, 16.5]	0.0%	—
<code>delivery_failure</code>	3.7%	[2.2, 6.2]	0.0%	(unreachable)
<code>hallucination</code>	3.5%	[2.0, 5.8]	2.4%	[0.7, 8.5]
<code>escalation_failure</code>	2.9%	[1.6, 5.2]	23.2%	[15.4, 33.4]
<code>premature_exit</code>	2.4%	[1.3, 4.5]	12.2%	[6.8, 21.0]

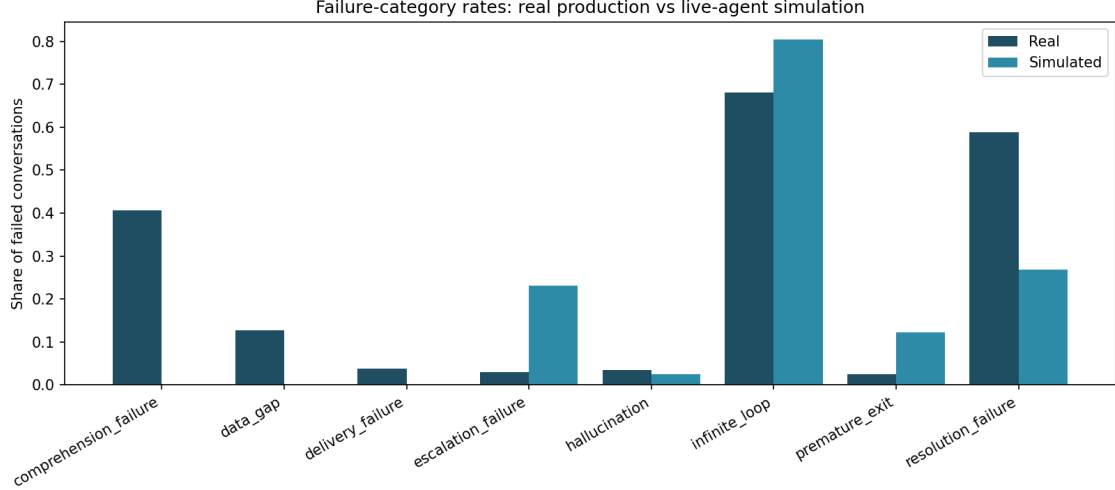


Figure 1: Failure-category rates among failed conversations: real production vs live-agent simulation, identical scorer, shared taxonomy.

4.4 Distributional divergence

On the reachable support, $\text{JSD}(\text{real}, \text{sim}) = \mathbf{0.2392}$ bits (95% CI [0.2010, 0.2979]) — roughly $48\times$ the split-half sampling noise floor of the real corpus (0.0050), and comparable to the real-vs-uniform anchor (0.2014). The honest reading: at the level of *which* failures occur, correspondence is strong (5/7, matching dominant mode); at the level of the full *distribution*, the two zero categories dominate the divergence and keep it far above noise. Including the unreachable category changes little ($\text{JSD} = 0.2470$).

4.5 Case-level correspondence

Frequency matching can be coincidental; mechanism matching is stronger evidence. The clearest matched pair observed: in a simulated test in which the customer disputes information on file, the agent repeated the *same* phone-number deflection message verbatim for 18 consecutive turns until the persona abandoned — a live reproduction of the mechanism behind production’s second-largest failure signal (`loop_stall`, 256 real failures), where the agent cycles rather than progresses when it has no path for the customer’s actual need. All three simulated hard failures in the 200-conversation run were of this mechanism.

5 Results: the scale study

Table 4 and Figure 2 present the seven independent batches and the cumulative pool.

Table 4: Scale study: independent batches (top) and cumulative pooling (bottom). “conv-min” is summed conversation time; wall time $\approx$ conv-min / 6 workers.

N	Failures	Fail rate	Cats	Coverage	JSD vs real (95% CI)	conv-min	Persona \$
10	3	30.0%	2	29%	0.213 (0.180–0.502)	3	0.05
50	12	24.0%	5	71%	0.254 (0.192–0.452)	10	0.28
100	24	24.0%	5	71%	0.232 (0.194–0.330)	22	0.64
200	43	21.5%	4	57%	0.247 (0.207–0.319)	38	0.96
400	90	22.5%	5	71%	0.248 (0.213–0.293)	87	2.40
800	219	27.4%	5	71%	0.225 (0.201–0.259)	173	5.04
1000	232	23.2%	5	71%	0.242 (0.217–0.276)	225	6.47

<i>Cumulative pooled</i>							
60	15	—	5	71%	0.210		
360	82	—	5	71%	0.225		
1560	391	—	5	71%	0.230		
2560	623	—	5	71%	0.233		

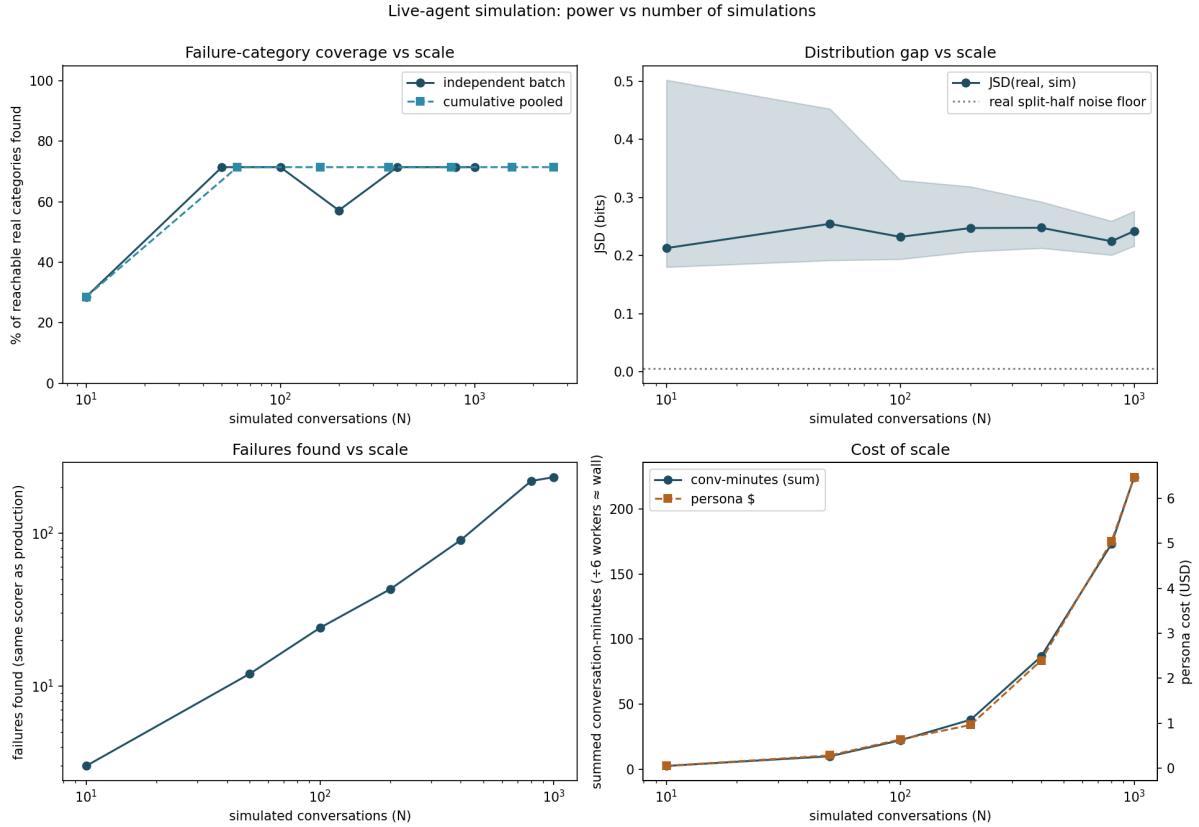


Figure 2: Simulation power vs number of simulations ($\log x$). Top-left: category coverage saturates at $N \approx 50$ (independent and cumulative). Top-right: JSD to the real distribution is scale-invariant; confidence intervals tighten with N . Bottom-left: failures found grow linearly. Bottom-right: cost of scale.

Three findings:

1. **Coverage saturates at $N \approx 50$.** A batch of ten finds 2 categories (29%); a batch of fifty finds 5 (71%); one thousand finds the same 5. The cumulative pool of 2,560 conversations

— $50\times$ the saturation point — still finds exactly 5. (The $N=200$ batch’s dip to 4 is sampling variance in the rare **hallucination** category, whose per-conversation probability is $\sim 1\%$.)

2. **Divergence is scale-invariant; precision is not.** The JSD point estimate stays in $[0.21, 0.25]$ at every scale, while its CI narrows from ± 0.16 ($N=10$) to ± 0.03 ($N=1000$). Scale buys certainty about the gap, not a smaller gap.
3. **Failure volume is linear and cheap.** Failures grow proportionally (3 at $N=10$; 232 at $N=1000$) at a stable 21–30% rate; full coverage costs roughly two wall-clock minutes and \$0.28 of persona traffic ($N=50$); a thousand-conversation batch costs ≈ 38 wall-minutes and \$6.47.

6 Discussion

Simulation power is gated by design, not scale. The saturation curve is the report’s core message. Everything the simulator *can* find, it finds almost immediately; what it cannot find at $N=50$ it still cannot find at $N=2,560$. The two absent categories jointly appear in over half of real failures, so closing them is the highest-leverage improvement available — and neither closes by running more simulations.

Why comprehension failures never appear. The scorer’s reachable comprehension signal is verbatim customer self-repetition. Real customers repeat themselves — 376 real conversations contain verbatim repeats — because they are frustrated, ignored, or misunderstood. LLM personas rephrase; across all 2,950 simulated conversations, not one repeated itself verbatim twice. This is the over-cooperativity phenomenon reported in the user-simulation literature, here confirmed on production data with a pre-registered prediction. The remedy is persona design (explicit repetition/frustration behaviours), not sampling temperature: published ablations show temperature moves surface diversity, not behavioural fidelity.

Why data-gap failures never appear. The fake customer was built to be *complete* — every order, warranty state, and CRM row present — because the simulation’s original purpose was to exercise agent logic. Real backends are incomplete (missing GSPN rows, stale statuses: 48 real failures). A test database that is too perfect makes an entire failure class invisible. The remedy is seeding *imperfect* fixtures: missing orders, null warranty fields, contradictory statuses.

What the simulation over-produces. `escalation_failure` (frustration without hand-off) and `premature_exit` are over-represented relative to production. Both excesses trace to adversarial suite composition: the generated tests deliberately probe refusal and abandonment boundaries that real traffic rarely exercises. This is the designed purpose of adversarial testing — finding failures production has not yet surfaced — but it means simulated failure *frequencies* should not be read as production forecasts without reweighting.

Implications. For practitioners: a small live-agent simulation ($N \approx 50$, minutes, cents) delivers the full coverage the current simulator design supports; invest additional budget in simulator realism, not repetition. For the dissertation: the behavioural result sharpens the static-mode RQ1 finding — even with the real agent in the loop, synthetic testing sees a strict subset of production failure kinds, and the boundary of that subset is now characterised mechanistically.

7 Threats to validity

1. **Rule-based ground truth.** Failure labels come from structured signals; the human agreement pass (Cohen’s κ on a 50-conversation blind packet) is prepared but not yet completed. An LLM pilot (lenient agreement 0.86, κ 0.51) suggests the criterion captures “customer-unserved friction” rather than only catastrophic failure. Headline claims should be finalised after the human pass.
2. **Oracle independence.** The comparison deliberately ignores the platform’s tool-signature test oracles (which mark 94–100% of these simulated conversations as “passed”); prior work in this project demonstrated such oracles miss semantic failures. Scorer parity avoids that bias but inherits the scorer’s blind spots symmetrically.
3. **Suite scope.** All batches draw from one 150-test generated suite (with seeded resampling above 150). Saturation at 5 categories is therefore a joint property of simulator *and* suite; a differently generated suite (e.g. production-feedback-seeded) could shift the plateau — this is the planned arm $\times$ context experiment.
4. **Single seeded customer.** The fake database bounds reachable data-gap and multi-customer failure space (pre-registered; observed).
5. **Clean transport.** Simulated tool calls never time out or rate-limit; chaos injection covers only part of this gap, and WhatsApp delivery failure is structurally absent (pre-registered as unreachable).
6. **Anonymisation noise.** Entity scrubbing may weaken fine-grained text signals on the real side (repetition detection is exact-match after normalisation and is robust to it).

8 Conclusion and future work

Against a verbatim copy of a production agent, LLM-persona simulation reproduces the majority (5/7) of reachable real failure categories — including the dominant loop/stall mode with a mechanistically identical case — at a production-like failure rate, and does so within ~ 50 conversations. What it misses, it misses structurally: over-cooperative personas never produce comprehension-failure signals, and a too-perfect test database never produces data-gap failures; 2,560 conversations do not buy either back. The distributional gap (JSD ≈ 0.24 vs a 0.005 noise floor) is therefore a design measurement, not a sampling artefact.

Future work, in leverage order. (1) *Frustrated/repetitive personas*: add explicit repetition and frustration behaviours to persona generation, then re-measure — predicted to open **comprehension** and grow **missed_escalation** realism. (2) *Imperfect data seeding*: fixture variants with missing/contradictory backend rows — predicted to open **data_gap**. (3) *Arm $\times$ context* 2×2 : blind vs production-feedback-seeded suites, with and without customer context, $N=200$ each — ties this behavioural analysis to the dissertation’s RQ3/RQ4 feedback result. (4) *Human annotation pass* to finalise ground-truth validity. (5) *Semantic oracles* (X1/X2): the same simulated corpora already contain semantic failures invisible to both the tool-signature oracles and the rule-based scorer (e.g. partially successful impersonation probes); LLM-judged categories, validated MAST-style against a human-annotated subsample, would extend the comparison beyond process signals.

References

- [1] *Lost in Simulation: LLM-Simulated Users are Unreliable Proxies for Human Users in Agentic Evaluations*, arXiv:2601.17087, 2026.

- [2] *Synthetic Users, Real Differences: an Evaluation Framework for User Simulation in Multi-Turn Conversations*, arXiv:2605.02624, 2026.
- [3] *RealUserSim: Bridging the Reality Gap in Agent Benchmarking via Grounded User Simulation*, arXiv:2605.20204, 2026.
- [4] *Mind the Sim2Real Gap in User Simulation for Agentic Tasks*, arXiv:2603.11245, 2026.
- [5] *The Sim-to-Real Gap of Foundation Model Agents: A Unified MDP Perspective*, arXiv:2606.07017, 2026.
- [6] M. Cemri, M. Z. Pan, S. Yang, et al., *Why Do Multi-Agent LLM Systems Fail?*, arXiv:2503.13657, 2025.
- [7] *What Would GPT Click: Practical Effects of Human-AI Behavioral Misalignment and the Cost of Synthetic Participants in User Experience*, arXiv:2605.18302, 2026.
- [8] *Statistical Properties and Power Analysis of Divergence Measures for Credit Risk Model Monitoring*, arXiv:2607.12407, 2026.
- [9] *Why AI Agents Break in Production: Failure Patterns and How to Detect Them*, Latitude Engineering Blog, 2026.

A Reproducibility

All artefacts live in the `investigation/` package accompanying this report: the anonymised real corpus (`02_data/real/`), every simulated corpus as consolidated `conversations.json` exports (`02_data/simulated/`), including the seven scale batches with run reports and failure inboxes), the three analysis scripts (`03_analysis_scripts/`), and machine-readable results (`04_results/`). The comparison regenerates with:

```
python3 compare_real_vs_sim.py --real samsung-conversations-anonymized.json \
--sim <conversations.json> [--sim ...] -o <outdir>
python3 scale_curves.py --real samsung-conversations-anonymized.json \
--batches results_scale_study -o <outdir>
```

Scale-study suites regenerate byte-identically from seeds ($42+N$) via `gen_scale_suites.py`. The raw (non-anonymised) export is deliberately excluded from this package.